Assessment Tool - 1

CO-2

Analytical Problem solving

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Course Code :- ITA0612

Course Name :- Machine learning

1) Perceptron model – Student Pass or Fail.

Sol:- Given

Study hours $(x_1) = 2$, weight $(w_1) = 0.6$
Attendance $(x_2) = 1$, weight $(w_2) = 0.4$
Bias $(b) = -0.5$

Formula:-

$$\text{Net Input} = w_1x_1 + w_2x_2 + b$$

$$= 0.6(2) + 0.4(1) - 0.5$$

$$= 1.1$$

Step Activation Function:-
If net input ≥ 0, output $= 1$
If net input < 0, output $= 0$
since $1.1 \geq 0$, the output is 1

Conclusion:- The student passes.

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2) weight update using the Delta Rule.

Given:-

Initial weight $= 0.5$
learning rate $(\eta) = 0.1$
Input $(x) = 2$
Actual output $(y) = 0.6$
Target output $(t) = 1$.

Error:-

$e = t - y$

$= 1 - 0.6$

$\boxed{e = 0.4}$

weight update:-

$\Delta w = \eta e x$

$= 0.1 \times 0.4 \times 2$

$\boxed{\Delta w = 0.08}$

new weight:-

$w_{new} = w + \Delta w$

$= 0.5 + 0.08$

$\boxed{w_{new} = 0.58}$

Conclusion:- The updated weight after one iteration is 0.58.

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3) Confusion matrix - Disease Prediction.

Given,

$TP = 180$

$TN = 150$

$FP = 20$

$FN = 50$

Total observation:-

$N = 180 + 150 + 20 + 50 = 400$

Accuracy:-

$= \frac{TP + TN}{N} \times 100$

$= \frac{180 + 150}{400} \times 100$

$= 82.5\%$

Precision:-

$$= \frac{TP}{TP+FP} \times 100$$

$$= \frac{180}{200} \times 100$$

$$= 90\%$$

Sensitivity (Recall):-

$$= \frac{TP}{TP+FN} \times 100$$

$$= \frac{180}{230} \times 100$$

$$= 78.26\%$$

Specificity:-

$$= \frac{TN}{TN+FP} \times 100$$

$$= \frac{150}{170} \times 100$$

$$= 88.24\%$$

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4) Genetic Algorithm - Maximize:-

$f(x) = x^2$

Given Population: $\{2, 4, 6, 8\}$

fitness values:-

Individual (x)	Fitness (x^2)
2	4
4	16
6	36
8	64

The best candidates:-

The best individuals are 8 and 6, with fitness values 64 and 36

Crossover operation:-

Binary representation

- 8 = 1000
- 6 = 0110

Conclusion:- Individual 8 is the best candidate, with maximum fitness of 64.

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5) Training and validation Accuracy.

Given:-

→ Training accuracy increases from 60% to 95%.

→ validation accuracy increases from 58%, to 72%, then decreases to 68%.

Issue : overfitting :-

Overfitting occurs when a neural network learns the training data too closely but performs poorly on unseen data.

Reasons:-

→ the model learns noise and unnecessary details from training data.

→ the model may be too complex for the available dataset.

Two techniques to Prevent overfitting:-

1. DroPout:- Randomly disables some neurons during training to reduce overdePendence on Particular neurons.

2. Early stopping:- stoPs training when validation Performance begins to deteriorate.

Importance of validation accuracy:-

→ Measure Performance on unseen validation data.
→ HelPs identify overfitting.
→ HelPs select the best model and training ePoch.

conclusion:- the model shows overfitting because training accuracy increases while validation accuracy decreases. DroPout and early stoPPing can improve generalization.